

Meningitis — Organisms by Age & Host





Swaddled neonate (gold star)

WHAT YOU SEE

Tiny swaddled newborn on the floor with a gold star — the neonatal age band where GBS, E. coli and Listeria reign.

WHAT IT ENCODES

Neonatal meningitis (<1 month) is caused by Group B Strep (*S. agalactiae*), *E. coli*, and *Listeria*. Empiric therapy is ampicillin PLUS cefotaxime (or ampicillin + gentamicin) — NOT ceftriaxone, which displaces bilirubin and risks kernicterus. GBS is acquired from maternal vaginal colonization; intrapartum penicillin after 35-37wk screening prevents early-onset disease.



Sausage baby in bassinet (GBS)

WHAT YOU SEE

Pink sausage-link 'baby' lying in the wicker bassinet — Group B Strep, the top neonatal pathogen.

WHAT IT ENCODES

Group B Streptococcus (*S. agalactiae*) is the leading cause of NEONATAL meningitis (early-onset from maternal colonization) and is rising in older adults with diabetes/cirrhosis. It is a gram-positive coccus in chains, beta-hemolytic, CAMP-test and hippurate positive. Intrapartum GBS prophylaxis prevents early- but not late-onset neonatal disease.



Crossed-out teddy bear (Hib, vaccine)

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WHAT YOU SEE

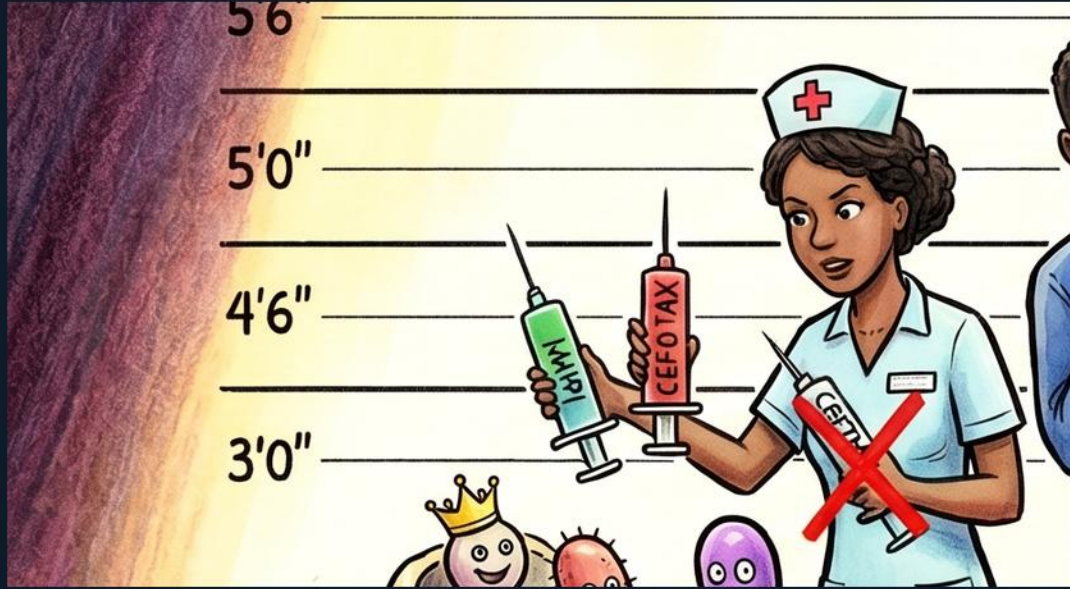
Red-X over the child's teddy bear — *Haemophilus influenzae* type b, now rare thanks to the conjugate vaccine.

WHAT IT ENCODES

Hib was historically a top cause of childhood meningitis but is now RARE in vaccinated children due to the conjugate vaccine. Consider it in unvaccinated/under-vaccinated kids, and in adults with CSF leak, sinusitis/otitis, or asplenia. It is a small gram-negative coccobacillus needing factors X and V.

THE BOARD TRAP

"complete vaccine sticker rules out Hib" -> wrong; verify TRUE immunization status — unvaccinated/partially vaccinated kids still get Hib, and non-type-b strains aren't vaccine-covered.



Nurse with crossed-out syringe

WHAT YOU SEE

Nurse holding a syringe marked with a red X over the bassinet — the neonatal 'don't use ceftriaxone' caution.

WHAT IT ENCODES

Ceftriaxone is **AVOIDED** in neonates: it displaces bilirubin (kernicterus) and precipitates with calcium-containing IV fluids. Use cefotaxime, or ampicillin + gentamicin, for the newborn instead. The neonatal pathogen trio (GBS, E. coli, Listeria) all need ampicillin on board.

THE BOARD TRAP

"ceftriaxone for the neonate" -> wrong; causes kernicterus and calcium precipitation — substitute cefotaxime or ampicillin + gentamicin.



Pneumococcus suspect (rash teen + diplococci)

WHAT YOU SEE

Teen lineup figure with the dark gram-positive diplococcus-shaped lollipop crossed out above — *S. pneumoniae*, the #1 adult cause.

WHAT IT ENCODES

Streptococcus pneumoniae is the #1 cause of bacterial meningitis in adults/elderly — gram-positive, lancet-shaped diplococci, alpha-hemolytic, optochin-sensitive, bile-soluble. It carries the HIGHEST case-fatality (~20-30%) with frequent sequelae. Empiric coverage is ceftriaxone PLUS vancomycin (resistant strains); dexamethasone has its strongest benefit here.

THE BOARD TRAP

"*N. meningitidis* is the most common adult cause" -> wrong; meningococcus dominates outbreaks/young adults, but pneumococcus is the single most common cause overall in adults.



Pneumococcus risk: alcohol & asplenia

WHAT YOU SEE

Bottle of alcohol and the basilar-skull/spleen icons beside the elderly man — the pneumococcal risk-factor cluster.

WHAT IT ENCODES

Pneumococcal risk factors: basilar skull fracture/CSF leak (recurrent meningitis), alcoholism, ASPLenia/sickle cell, diabetes, multiple myeloma/hypogammaglobulinemia, cochlear implants. Asplenic/sickle patients are vulnerable to ALL encapsulated bugs (pneumococcus, meningococcus, Hib) and must be vaccinated.

THE BOARD TRAP

"recurrent pneumococcal meningitis = just re-treat" -> wrong; recurrence signals an anatomic CSF leak (basilar fracture, cribriform defect) or complement/Ig deficiency — hunt for it.



Detective in fedora (*N. meningitidis*)

WHAT YOU SEE

Purple gangster in trench coat and fedora holding playing cards — *Neisseria meningitidis*, the young-adult/dorm outbreak suspect.

WHAT IT ENCODES

Neisseria meningitidis leads in ADOLESCENTS/young adults, especially crowded settings (dorms, barracks) — gram-negative diplococci that ferment maltose AND glucose, grow on Thayer-Martin. Serogroups B/C/W/Y predominate; MenACWY and MenB are recommended for adolescents, college freshmen in dorms, recruits, and asplenic/complement-deficient patients. Endotoxin drives DIC and Waterhouse-Friderichsen (adrenal hemorrhage, shock).

THE BOARD TRAP

"petechial rash + fever = benign viral exanthem" -> wrong; a non-blanching petechial/purpuric rash with fever and meningismus is meningococcemia until proven otherwise — immediate antibiotics, droplet isolation, contact prophylaxis.



Complement deficiency (skull/MAC)

WHAT YOU SEE

Bare skull at the elderly nurse's feet — the membrane-attack-complex (terminal complement) defect that invites recurrent *Neisseria*.

WHAT IT ENCODES

Terminal complement deficiency (C5-C9, the MAC) and complement inhibitors (eculizumab, ravulizumab) markedly predispose to recurrent/invasive *Neisseria*, because MAC is required to lyse *Neisseria*. These patients need meningococcal vaccination and sometimes prophylactic antibiotics.

THE BOARD TRAP

"don't work up immune deficiency after recurrent *Neisseria*" -> wrong; recurrent meningococcal disease should trigger a CH50/complement workup for terminal complement deficiency.



Listeria suspect (purple rod villain)

WHAT YOU SEE

Long purple rod-shaped character behind the old man — *Listeria monocytogenes*, the gram-positive ROD.

WHAT IT ENCODES

Listeria monocytogenes is a gram-positive ROD with tumbling motility and actin-based 'rocket' propulsion, causing meningitis in neonates, adults >50, pregnant women, and the immunocompromised; it favors the brainstem (rhombencephalitis). It is intrinsically resistant to ALL cephalosporins. Treat with ampicillin (+ gentamicin for synergy), or TMP-SMX if penicillin-allergic.

THE BOARD TRAP

"ceftriaxone + vancomycin covers everything" -> wrong; cephalosporins have ZERO *Listeria* activity — failing to ADD ampicillin in a *Listeria*-risk host is a fatal omission.



Listeria food: deli meat & soft cheese

WHAT YOU SEE

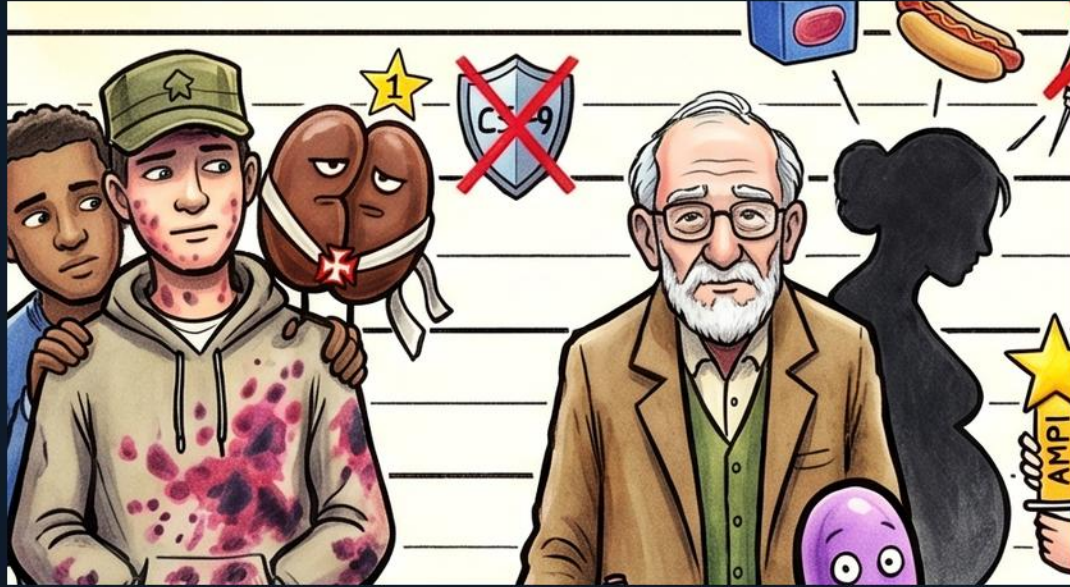
Hot dog and wedge of cheese floating near the cooler/lunchbox — the cold-tolerant deli-meat and soft-cheese sources of Listeria.

WHAT IT ENCODES

Listeria is foodborne via deli/processed meats, unpasteurized soft cheeses, hot dogs, pate, smoked seafood, and contaminated coleslaw/produce. It uniquely multiplies at refrigeration temperatures (cold enrichment, 4°C). Pregnant women avoid these foods due to transplacental infection, miscarriage, and neonatal listeriosis (granulomatosis infantiseptica).

THE BOARD TRAP

"it was refrigerated, so not Listeria" -> wrong; Listeria grows at fridge temperatures — 'kept cold' is no reassurance, the cold-deli history is a positive clue.



Listeria host: elderly + immunocompromised

WHAT YOU SEE

The gray-bearded old man with a cane in the lineup center — the age >50 / impaired-cell-mediated-immunity Listeria host.

WHAT IT ENCODES

Add ampicillin for Listeria in the classic hosts: <1 month (neonate), >50 years, pregnancy, and impaired cell-mediated immunity (transplant, chronic steroids, TNF inhibitors, HIV, malignancy). Listeria is intracellular and controlled by T cells, so any cell-mediated defect raises risk.

THE BOARD TRAP

"only neonates and the very elderly need Listeria coverage" -> wrong; pregnant women and any cell-mediated-immunity defect (steroids, transplant, HIV) are equally at risk regardless of age.



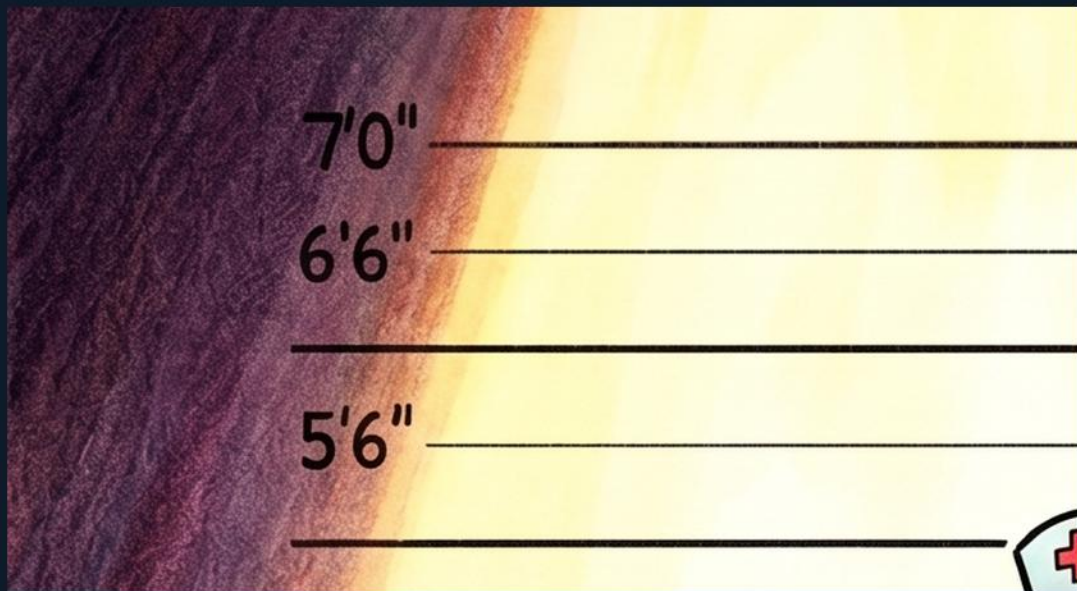
Cooler / lunchbox (refrigeration cue)

WHAT YOU SEE

The open insulated cooler at the old man's feet — refrigerated food, reinforcing *Listeria*'s cold-tolerance.

WHAT IT ENCODES

The cooler reinforces that *Listeria* survives and grows at 4°C, which is why refrigeration does not exclude it and why outbreaks trace to chilled ready-to-eat foods. Empiric ampicillin must accompany ceftriaxone + vancomycin whenever the CSF Gram stain shows a gram-positive ROD or the host is high-risk.



Mortality ranking (height chart)

WHAT YOU SEE

The police-lineup height ruler running up the left wall — a stand-in for the case-fatality ranking of the suspects.

WHAT IT ENCODES

Approximate case-fatality ranking: *S. pneumoniae* ~20-30% (highest), *Listeria* ~15-20%, *H. influenzae* ~5%, *N. meningitidis* ~3-7% (lowest when treated). Mortality reflects host and virulence; pneumococcal survivors often have lasting neurologic deficits.

THE BOARD TRAP

"meningococcus is the deadliest because of its rash" -> wrong; treated meningococcal meningitis has comparatively LOW case fatality — pneumococcus and *Listeria* are deadlier per case.



Empiric regimen by age (bottom rule)

WHAT YOU SEE

The numbered evidence placard/vial cluster on the right pedestal — the age-based empiric antibiotic summary.

WHAT IT ENCODES

Age-based empiric therapy: NEONATES → ampicillin + cefotaxime (or + gentamicin); 1mo-50yr → ceftriaxone + vancomycin; >50yr or immunocompromised → ceftriaxone + vancomycin + AMPICILLIN. Bottom line: gram-positive RODS or a *Listeria*-risk host means ADD AMPICILLIN — cephalosporins miss *Listeria* entirely.

THE BOARD TRAP

"ceftriaxone + vancomycin alone for an elderly/immunocompromised patient" → wrong; this pair misses *Listeria* — omitting ampicillin in neonates, >50, pregnant, alcoholic, or immunocompromised is a classic fatal error.

Summary — every symbol

1. Swaddled neonate (gold star)

Neonatal meningitis (<1 month) is caused by Group B Strep (S)

2. Sausage baby in bassinet (GBS)

Group B Streptococcus (S)

3. Crossed-out teddy bear (Hib, vaccine)

Hib was historically a top cause of childhood meningitis but is now RARE in vaccinated children due to the conjugate vaccine

4. Nurse with crossed-out syringe

Ceftriaxone is AVOIDED in neonates: it displaces bilirubin (kernicterus) and precipitates with calcium-containing IV fluids

5. Pneumococcus suspect (rash teen + diplococci)

Streptococcus pneumoniae is the #1 cause of bacterial meningitis in adults/elderly — gram-positive, lancet-shaped diplococci, alpha-hemolytic, optochin-sensitive, bile-soluble

6. Pneumococcus risk: alcohol & asplenia

Pneumococcal risk factors: basilar skull fracture/CSF leak (recurrent meningitis), alcoholism, ASPLENIA/sickle cell, diabetes, multiple myeloma/hypogammaglobulinemia, cochlear implants

7. Detective in fedora (N. meningitidis)

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8. Complement deficiency (skull/MAC)

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9. Listeria suspect (purple rod villain)

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10. Listeria food: deli meat & soft cheese

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11. Listeria host: elderly + immunocompromised

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12. Cooler / lunchbox (refrigeration cue)

The cooler reinforces that Listeria survives and grows at 4°C, which is why refrigeration does not exclude it and why outbreaks trace to chilled ready-to-eat foods

13. Mortality ranking (height chart)

Approximate case-fatality ranking: S

14. Empiric regimen by age (bottom rule)

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